

ONE STANDARD DEVIATION DOES NOT SUFFICE: A LOWER BOUND FOR DISCREPANCY

ABSTRACT. In this short note, we prove that for each sufficiently large $n \in \mathbb{N}$ admitting an $n \times n$ Hadamard matrix, there exists $A = A_n \in \{\pm 1\}^{n \times n}$ such that

$$\text{disc}(A) \geq (1 + \delta)\sqrt{n}, \quad \delta = 10^{-20}, \quad \text{disc}(A) := \inf_{x \in \{\pm 1\}^n} \|Ax\|_\infty.$$

In particular, this implies that

$$\limsup_{n \rightarrow \infty} \sup_{A \in \{\pm 1\}^{n \times n}} \frac{1}{\sqrt{n}} \text{disc}(A) \geq 1 + \delta > 1,$$

thereby proving [BDL⁺26, Conjecture 12]. The crux of our construction is a probabilistic argument based on a random modification of Hadamard matrix. The main result of this paper were generated using GPT-6 Astra.

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1. INTRODUCTION AND MAIN RESULTS

Given $n \in \mathbb{N}$ and $A \in \mathbb{R}^{n \times n}$, we define the discrepancy of A as $\text{disc}(A) := \inf_{x \in \{\pm 1\}^n} \|Ax\|_\infty$. One of the motivations of this question is to understand how much smaller is $\text{disc}(A)$ compared to the value, e.g. in expectation, when x is drawn uniformly from the hypercube. In 1985, Joel Spencer [Spe85] showed the celebrated “Six Deviations Suffice” Theorem, which states that for any positive integer n and any $A \in \{\pm 1\}^{n \times n}$ we have $\text{disc}(A) \leq 6\sqrt{n}$. There have since been improvements on this constant, but the question we will pursue here concerns lower bounds. In fact, as discussed in [BDL⁺26], there seems to be no known construction (or proof of existence) of an infinite family achieving a lower bound strictly larger than one. This motivated them formulate the following conjecture (see Conjecture 12 therein).

Conjecture 1.1. *We have*

$$\limsup_{n \rightarrow \infty} \sup_{A \in \{\pm 1\}^{n \times n}} \frac{1}{\sqrt{n}} \text{disc}(A) \geq 1 + \delta > 1,$$

The main goal of this short note is to prove Conjecture 1.1. Recall that a matrix $H_n \in \{\pm 1\}^{n \times n}$ is a Hadamard matrix if $H_n^\top H_n = nI_n$.

Theorem 1.2. *There exists an absolute constant N_0 such that the following holds. For each even $n \geq N_0$ admitting an $n \times n$ Hadamard matrix H_n , there exists $A_n \in \{\pm 1\}^{n \times n}$ such that $\text{disc}(A_n) \geq (1 + \delta)\sqrt{n}$, where $\delta = 10^{-20} > 0$.*

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In particular, since it is well known that an $n \times n$ Hadamard matrix exists for all $n = 2^k, k \geq 1$, Theorem 1.2 immediately implies Conjecture 1.1.

Our proof is based on a probabilistic argument that randomly modifies the signs of the Hadamard matrix H_n . More precisely, given an $n \times n$ Hadamard matrix H , independently for each row i , we choose a uniform subset $S_i \subseteq [n]$ of cardinality $r = \lfloor \delta n \rfloor$. Set

$$A_{i,j} = \begin{cases} H_{i,j}, & j \notin S_i; \\ -H_{i,j}, & j \in S_i. \end{cases} \quad (1.1)$$

It suffices to show the following result.

Theorem 1.3. *For all sufficiently large n and every $n \times n$ Hadamard matrix H , the random matrix A defined in (1.1) with $\delta = 10^{-20}$ satisfies*

$$\mathbb{P}(\text{disc}(A) \leq (1 + \delta)\sqrt{n}) \leq e^{-n/1600} + e^{-(1-\log 2)n}.$$

It is clear that Theorem 1.3 implies Theorem 1.2.

Statement on AI use. The main results in this paper were generated by simply prompting GPT-6 Astra with Conjecture 1.1, without any mathematical input.

2. PROOF OF THE MAIN THEOREM

In this section, we prove Theorem 1.3. We first record a number of probabilistic and combinatorial inputs that will be used in the proof. The proof of those inputs are postponed to Section 3. We first record a bound for the number of “nearly flat” images of Hadamard matrices.

Lemma 2.1. *Let $H \in \{-1, 1\}^{n \times n}$ satisfy $H^\top H = nI$, where $n \geq 2$. Then*

$$\#\mathcal{F}(H) := \#\left\{x \in \{-1, 1\}^n : \left\| \frac{1}{\sqrt{n}} Hx \right\|_1 > \frac{9}{10}n\right\} \leq 2^n e^{-n/800}.$$

We then record two facts for a random r -subset $S \subseteq [n]$.

Lemma 2.2. *Let $1 \leq r \leq n$ be an integer, let $w_1, \dots, w_n \in \{-1, 1\}$, let S be a uniform r -subset of $[n]$, and let $B = \sum_{j \in S} w_j$. Then, for $u \geq 0$,*

$$\mathbb{P}(B - \mathbb{E}B \geq u) \leq e^{-u^2/(2r)}.$$

The same estimate holds for the lower tail.

Lemma 2.3. *Fix $\delta \in (0, 1)$ and a compact interval $\Gamma \subset \mathbb{R}$. Let $r = \lfloor \delta n \rfloor$, and suppose that $w \in \{-1, 1\}^n$ satisfies $\sum_j w_j = v\sqrt{n}$ with $v \in \Gamma$. Let S be a uniform r -subset of $[n]$, and put $Y = n^{-1/2}(\sum_j w_j - 2 \sum_{j \in S} w_j)$. There exists $\varepsilon_n = \varepsilon_n(\Gamma, \delta) \geq 0$ tending to zero such that, uniformly over all such w ,*

$$\sup_{T \in \mathbb{R}} \left| \mathbb{P}(Y \leq T) - \Phi\left(\frac{T - (1 - 2\delta)v}{2\sqrt{\delta(1 - \delta)}}\right) \right| \leq \varepsilon_n.$$

Here Φ is the standard normal distribution function.

For $v \in \mathbb{R}$ such that $v\sqrt{n} \in \{-n, -n + 2, \dots, n\}$, choose $w \in \{-1, 1\}^n$ with $\sum_j w_j = v\sqrt{n}$ and define

$$p_n(v) = \mathbb{P}\left(\left|\sum_j w_j - 2 \sum_{j \in S} w_j\right| \leq (1 + \delta)\sqrt{n}\right), \quad |S| = \lfloor \delta n \rfloor,$$

where S is uniformly chosen. This probability depends only on v , not on the particular choice of w .

Lemma 2.4. Set $\delta = 10^{-20}$, $\eta = 1/1600$, and $\lambda = 1/\sqrt{8\pi\delta}$. For all sufficiently large n and every $v \in \mathbb{R}$ satisfying $v\sqrt{n} \in \{-n, -n+2, \dots, n\}$, we have

$$p_n(v) \leq \exp(-\log 2 + \eta - \lambda(v^2 - 1)). \quad (2.1)$$

We now provide the proof of Theorem 1.3.

Proof of Theorem 1.3. Put $t = 1 + \delta$, $\delta_n = r/n$, and $a_n = 1 - 2\delta_n$, and take n to be sufficiently large that $r \geq 1$ and Lemma 2.4 applies. Thus $\delta_n \leq \delta$ and $a_n \geq 1 - 2\delta$. We use $\eta = 1/1600$ and $\lambda = 1/\sqrt{8\pi\delta}$ as in Lemma 2.4. We will apply a union bound to control $\|Ax\|_\infty$ for all $x \in \{\pm 1\}^n$. Fix a sign vector x and write $v = Hx/\sqrt{n}$. Orthogonality gives

$$\sum_i v_i^2 = n. \quad (2.2)$$

We divide into two cases:

Case 1: $x \in \mathcal{F}(H)$. The random choices in different rows of A are independent, so Lemma 2.4 and (2.2) imply the uniform bound

$$\begin{aligned} \mathbb{P}(\|Ax\|_\infty \leq t\sqrt{n}) &= \prod_i p_n(|v_i|) \\ &\leq \exp\left((- \log 2 + \eta)n - \lambda \sum_i (v_i^2 - 1)\right) = 2^{-n} e^{\eta n}. \end{aligned}$$

Combined with Lemma 2.1 and a union bound, this gives

$$\begin{aligned} \mathbb{P}(\|Ax\|_\infty \leq t\sqrt{n} \text{ for some } x \in \mathcal{F}(H)) \\ \leq 2^n e^{-n/800} \cdot 2^{-n} e^{\eta n} = e^{-n/1600}. \end{aligned} \quad (2.3)$$

Case 2: $x \notin \mathcal{F}(H)$. Then $\sum_i |v_i| \leq 0.9n$. Lemma 2.2 gives

$$\begin{aligned} \mathbb{P}(\|Ax\|_\infty \leq t\sqrt{n}) &= \prod_i p_n(|v_i|) \\ &\leq \prod_{i=1}^n \exp\left(-\frac{(a_n|v_i| - t)_+^2}{8\delta_n}\right). \end{aligned} \quad (2.4)$$

Set $u_i = (a_n|v_i| - t)_+$. For $w \geq 0$ one has $w^2 \leq tw + w(w - t)_+$. Consequently, by Cauchy-Schwarz and (2.2),

$$\begin{aligned} a_n^2 n &\leq t a_n \sum_i |v_i| + a_n \sum_i |v_i| u_i \\ &\leq \frac{9}{10} t a_n n + a_n \sqrt{n} \left(\sum_i u_i^2\right)^{1/2}. \end{aligned}$$

Since $a_n - 9t/10 \geq 1/10 - 29\delta/10 > 0$, division by $a_n \sqrt{n}$, rearrangement, and squaring yield

$$\sum_i u_i^2 \geq \left(a_n - \frac{9}{10}t\right)^2 n \geq \left(\frac{1}{10} - \frac{29}{10}\delta\right)^2 n.$$

Substituting this estimate into (2.4) and applying a union bound, we obtain

$$\begin{aligned} \mathbb{P}(\|Ax\|_\infty \leq t\sqrt{n} \text{ for some } x \notin \mathcal{F}(H)) \\ \leq 2^n \exp\left(-\frac{(1/10 - (29/10)\delta)^2 n}{8\delta}\right) \leq e^{-(1-\log 2)n}. \end{aligned} \quad (2.5)$$

Combining (2.3) and (2.5) proves the result. $\square$

3. PRELIMINARY INPUTS

In this section, we complete the omitted proofs in Lemmas 2.1–2.4.

3.1. Proof of Lemma 2.1.

Proof. Let $Q = H/\sqrt{n}$, let X be uniform on the sign cube, and put $F(x) = \|Qx\|_1$. Orthogonality of any two columns of H implies that n is even. Write $n = 2m$. An elementary binomial summation, together with $\binom{2m}{m}/4^m \leq 1/\sqrt{\pi m}$, gives

$$\mathbb{E}F(X) = \sqrt{n} \mathbb{E} \left| \sum_{j=1}^n X_j \right| = \sqrt{n} 2m \frac{\binom{2m}{m}}{4^m} \leq \sqrt{\frac{2}{\pi}} n < \frac{4}{5}n.$$

We next prove the concentration estimate needed here. Denote by $x^{(j)}$ the vector obtained by reversing the j th coordinate of x . Choose $y_i \in \{-1, 1\}$ such that $y_i(Qx)_i = |(Qx)_i|$; either sign may be used when $(Qx)_i = 0$. The vector $g = Q^\top y$ is a subgradient of F at x , and $\|g\|_2^2 = n$. Convexity therefore gives

$$F(x) - F(x^{(j)}) \leq 2x_j g_j, \quad \sum_{j=1}^n (F(x) - F(x^{(j)}))_+^2 \leq 4n.$$

Talagrand's concentration inequality then implies that

$$\mathbb{P}(F(X) \geq \mathbb{E}F(X) + u) \leq e^{-u^2/(8n)}.$$

Taking $u = n/10$ proves the claimed counting bound. $\square$

3.2. Proof of Lemma 2.2.

Proof. If $r = n$, then B is deterministic and the conclusion is immediate. Suppose that $r < n$. Generate S by sampling distinct indices $J_1, \dots, J_r$ uniformly without replacement. Let $\mathcal{G}_k = \sigma(J_1, \dots, J_k)$ and $M_k = \mathbb{E}[B \mid \mathcal{G}_k]$, so that $M_0 = \mathbb{E}B$ and $M_r = B$. For $1 \leq k \leq r$, let $m_{k-1} = (n-k+1)^{-1} \sum_{j \notin \{J_1, \dots, J_{k-1}\}} w_j$ be the mean of the remaining signs. A direct computation gives $M_k - M_{k-1} = (n-r)(w_{J_k} - m_{k-1})/(n-k)$. Conditionally on $\mathcal{G}_{k-1}$, this increment has mean zero and is supported on an interval of length at most $2(n-r)/(n-k) \leq 2$.

For any centered random variable Z supported on an interval of length at most two, one has $\mathbb{E}e^{\theta Z} \leq e^{\theta^2/2}$ for every $\theta \in \mathbb{R}$. Indeed, the second derivative of $\log \mathbb{E}e^{\theta Z}$ is the variance under the corresponding exponentially tilted law, which is at most one because its support remains in the same interval. Integrating twice and using $\mathbb{E}Z = 0$ proves the bound. Applying this conditionally to the martingale increments and iterating gives $\mathbb{E}e^{\theta(B - \mathbb{E}B)} \leq e^{r\theta^2/2}$. For $u > 0$, Markov's inequality with $\theta = u/r$ yields

$$\mathbb{P}(B - \mathbb{E}B \geq u) \leq \exp\left(-\theta u + \frac{r\theta^2}{2}\right) = \exp\left(-\frac{u^2}{2r}\right).$$

The case $u = 0$ is immediate. Replacing every w_j by $-w_j$ gives the lower-tail estimate. $\square$

3.3. Proof of Lemma 2.3.

Proof. Let $K = \#\{j : w_j = 1\} = (n + v\sqrt{n})/2$, put $p = K/n$ and $\delta_n = r/n$, and let $X = \#\{j \in S : w_j = 1\}$. Then $p = 1/2 + v/(2\sqrt{n})$ and

$$\mathbb{P}(X = k) = \frac{\binom{K}{k} \binom{n-K}{r-k}}{\binom{n}{r}}, \quad Y = (1 - 2\delta_n)v - \frac{4}{\sqrt{n}}(X - rp).$$

Throughout the proof, all error estimates are uniform over the allowed $v \in \Gamma$. Since Γ is compact, $p = 1/2 + O_\Gamma(n^{-1/2})$, while $\delta_n \rightarrow \delta$. In particular, p and δ_n stay bounded away from zero and one for all sufficiently large n .

Put $\tau_n^2 = n\delta_n(1 - \delta_n)p(1 - p)$. We first establish a uniform local normal approximation. Fix $M > 0$ and consider integers k with $|k - rp| \leq M\sqrt{n}$. Writing $u = k - rp$, Stirling's formula gives

$$\begin{aligned} \log \mathbb{P}(X = k) &= Kh \left(\delta_n + \frac{u}{K} \right) + (n - K)h \left(\delta_n - \frac{u}{n - K} \right) - nh(\delta_n) \\ &\quad - \frac{1}{2} \log(2\pi\tau_n^2) + o(1), \end{aligned}$$

where $h(s) = -s \log s - (1-s) \log(1-s)$. The error is uniform for the indicated k and v . Taylor expansion about δ_n cancels the linear terms, and $h''(s) = -1/[s(1-s)]$ gives

$$\begin{aligned} Kh \left(\delta_n + \frac{u}{K} \right) + (n - K)h \left(\delta_n - \frac{u}{n - K} \right) - nh(\delta_n) \\ = -\frac{u^2}{2\tau_n^2} + O_{\Gamma, \delta, M}(n^{-1/2}). \end{aligned}$$

Consequently,

$$\mathbb{P}(X = k) = \frac{1 + o(1)}{\sqrt{2\pi} \tau_n} \exp \left(-\frac{(k - rp)^2}{2\tau_n^2} \right), \quad |k - rp| \leq M\sqrt{n},$$

uniformly over k and v .

The sampling indicators give $\mathbb{E}X = rp$ and $\text{Var}(X) = rp(1 - p)(n - r)/(n - 1) = \tau_n^2 n/(n - 1)$. Thus Chebyshev's inequality bounds $\mathbb{P}(|X - rp| > M\sqrt{n})$ by $C_{\Gamma, \delta}/M^2$, uniformly in v . Summing the local approximation over $|k - rp| \leq M\sqrt{n}$ and comparing with the corresponding Gaussian Riemann sums, uniformly in the upper endpoint, therefore yields

$$\sup_{z \in \mathbb{R}} \left| \mathbb{P} \left(\frac{X - rp}{\tau_n} \leq z \right) - \Phi(z) \right| \rightarrow 0,$$

uniformly over all allowed v . Here one first lets $n \rightarrow \infty$ for fixed M and then lets $M \rightarrow \infty$; the Gaussian tails also vanish uniformly because $\tau_n/\sqrt{n}$ stays bounded above and away from zero.

Finally, uniformly over $v \in \Gamma$,

$$(1 - 2\delta_n)v \rightarrow (1 - 2\delta)v, \quad \frac{4\tau_n}{\sqrt{n}} = 4\sqrt{\delta_n(1 - \delta_n)p(1 - p)} \rightarrow 2\sqrt{\delta(1 - \delta)}.$$

Using the representation of Y above, reflection symmetry of the standard normal distribution, and continuity of its distribution function proves the asserted uniform estimate. More explicitly, the uniform distribution-function approximation also holds for left limits, so reflecting $(X - rp)/\tau_n$ causes no endpoint problem. The convergence of the means and of the positive standard deviations then gives uniform convergence of the normal distribution functions for all $T \in \mathbb{R}$. $\square$

3.4. Proof of Lemma 2.4.

Proof. Replacing w by $-w$ shows that $p_n(-v) = p_n(v)$, so it is enough to consider $v \geq 0$. Put $t = 1 + \delta$, $a = 1 - 2\delta$, and $\sigma = 2\sqrt{\delta(1 - \delta)}$. Also write $\delta_n = \lfloor \delta n \rfloor / n$ and $a_n = 1 - 2\delta_n$, and take n sufficiently large that $\lfloor \delta n \rfloor \geq 1$. Put $B = \sum_{j \in S} w_j$ and $Y = v - 2B/\sqrt{n}$. Then $\mathbb{E}B = \delta_n v \sqrt{n}$ and $Y = a_n v - 2(B - \mathbb{E}B)/\sqrt{n}$. Since $\{|Y| \leq t\} \subseteq \{Y \leq t\}$, Lemma 2.2 gives

$$p_n(v) \leq \exp\left(-\frac{(a_n v - t)_+^2}{8\delta_n}\right) \leq \exp\left(-\frac{(av - t)_+^2}{8\delta}\right),$$

where the second inequality uses $a_n \geq a$ and $\delta_n \leq \delta$.

We also use the elementary Gaussian inequality $\log \Phi(z) \leq -\log 2 + \sqrt{2/\pi} z$ for every $z \in \mathbb{R}$. To see this, let $\varphi(z) = (2\pi)^{-1/2} e^{-z^2/2}$ and $R(z) = \varphi(z)/\Phi(z)$. Differentiation gives $R'(z) = -R(z)(z + R(z)) < 0$, since $z\Phi(z) + \varphi(z) = \int_{-\infty}^z (z - u)\varphi(u) du > 0$. Thus $\log \Phi$ is concave, and its tangent at zero is the claimed upper bound.

Case 1: $0 \leq v \leq 1 - 10\sqrt{\delta}$. In this range, $\lambda(1 - v^2) \geq (20 - 100\sqrt{\delta})/\sqrt{8\pi} > \log 2$. The right-hand side of (2.1) is therefore greater than one, and the conclusion follows from $p_n(v) \leq 1$.

Case 2: $|v - 1| \leq 10\sqrt{\delta}$. Apply Lemma 2.3 with the fixed compact interval $\Gamma = [1 - 10\sqrt{\delta}, 1 + 10\sqrt{\delta}]$. Uniformly over the allowed v in this interval,

$$p_n(v) \leq \mathbb{P}(Y \leq t) \leq \Phi\left(\frac{t - av}{\sigma}\right) + \varepsilon_n, \quad \varepsilon_n \rightarrow 0.$$

The Gaussian inequality and the choice $\lambda = 1/\sqrt{8\pi\delta}$ give

$$\begin{aligned} \log \Phi\left(\frac{t - av}{\sigma}\right) + \log 2 + \lambda(v^2 - 1) &\leq \sqrt{\frac{2}{\pi}} \frac{t - av}{\sigma} + \lambda(v^2 - 1) \\ &= \lambda(v - 1)^2 + \frac{t - av - \sqrt{1 - \delta}(1 - v)}{\sqrt{2\pi\delta(1 - \delta)}}. \end{aligned}$$

The numerator in the last fraction equals $1 + \delta - \sqrt{1 - \delta} + v(\sqrt{1 - \delta} - 1 + 2\delta)$, which lies between zero and $2\delta(1 + v) \leq 6\delta$. Moreover, $\lambda(v - 1)^2 \leq 25\sqrt{\delta}$ and $\sqrt{2\pi\delta(1 - \delta)} \geq 2\sqrt{\delta}$. The last display is consequently at most $30\sqrt{\delta} < \eta/2$. Hence

$$p_n(v) \leq \exp\left(-\log 2 + \frac{\eta}{2} - \lambda(v^2 - 1)\right) + \varepsilon_n.$$

Since $e^{-\log 2 - \lambda(v^2 - 1)} \geq e^{-6}$ throughout this interval, taking n so large that $\varepsilon_n \leq e^{-6}(e^\eta - e^{\eta/2})$ proves (2.1) uniformly in this case.

Case 3: $v \geq 1 + 10\sqrt{\delta}$. Put $u = v - 1 \geq 10\sqrt{\delta}$. Since $\delta = 10^{-20}$, one has $av - t = (1 - 2\delta)u - 3\delta \geq 3u/4$ and $\lambda \leq 1/(4\sqrt{\delta})$. Therefore

$$\begin{aligned} \frac{(av - t)^2}{8\delta} - \lambda(v^2 - 1) &\geq \frac{9u^2}{128\delta} - \frac{u^2 + 2u}{4\sqrt{\delta}} \\ &\geq \frac{u^2}{16\delta} - \frac{u}{2\sqrt{\delta}} \geq \frac{5}{4} > \log 2. \end{aligned}$$

The second inequality uses $\sqrt{\delta} \leq 1/32$, and the third uses $u/\sqrt{\delta} \geq 10$. The sampling tail bound at the start of the proof now gives $p_n(v) \leq \exp(-\log 2 - \lambda(v^2 - 1))$, which is stronger than (2.1). The three cases exhaust all nonnegative v , and symmetry completes the proof. $\square$

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